

SmartEdu Portal

EduScribe

Test Case Documentation

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Document date	September 26, 2026
Total test cases	102
Automated	82 (RLS 13, unit 31, component 23, end-to-end 15)
Manual	20
Executed and passing	54 (unit and component suites)
Awaiting first run	48 (see Section 5)

1. Testing strategy

The system is tested in four layers. Each layer catches a class of defect the others structurally cannot, and the reason for having four is that this project has already shipped one defect from each of the first three.

1.1 Unit and render tests (Vitest)

Pure functions — scoring, deadline formatting, teaching-load rules, retry behaviour — plus a render smoke test that paints every dashboard tab once and asserts only that it does not throw. That smoke test exists because a component once used an icon it had never imported: the production build passed and every test passed, because Vite does not resolve free identifiers, and the undeclared global reached every student as a blank screen.

1.2 Component interaction tests (Vitest + jsdom)

Rendering a screen proves nothing about whether its buttons work. These drive the two rebuilt admin screens through a real DOM and assert the rules they enforce — one teacher per subject per grade, the draft that writes nothing until Assign, the red fields. None of those rules is in the database, so when one breaks it breaks silently.

1.3 Row Level Security tests (SQL)

The application has no backend server. The browser talks directly to Supabase, which makes Row Level Security the only barrier between one student and another student's answers. No unit test touches a policy. This suite sets `request.jwt.claims` and asks the database questions as each user, comparing what comes back against what the policies promise.

1.4 End-to-end tests (Playwright)

A real browser, a real login, the real database. This is the only layer that exercises authentication, routing and the network together. It earns its cost: the worst performance defect found in this project — an 8.8 second deadlock on every login for every user — was invisible in the source and invisible to every other test, and was diagnosed from a browser console.

1.5 Manual test cases

Section 5 covers what the automated suites deliberately do not. There is no separate test database, so every automated suite is read-only; any flow that creates, distributes, answers or scores real work is verified by hand.

2. Test environment

Application	React 18, Vite 5, React Router
Backend	Supabase — PostgreSQL, Auth, Storage, auto-generated REST API, Row Level Security. No application server exists.
Unit / component	Vitest, jsdom, @testing-library/react — `npm test`
End-to-end	Playwright (Chromium) — `npm run test:e2e`
RLS	Supabase SQL Editor — <code>backend/database/tests/phase5-01-rls-tests.sql</code>
Test data	60 seeded students and 48 seeded teachers (<code>backend/database/seeds</code>), all on <code>example.com</code> addresses, deleted after testing.

Because there is no separate test instance, the end-to-end and RLS suites run against the live database and are written to be read-only. The RLS suite's two write probes roll themselves back.

3. Security test cases – Row Level Security

Source: backend/database/tests/phase5-01-rls-tests.sql. Each case runs as a real logged-in user. A result of SKIP means the fixture the case needs does not exist yet; it is not a pass, and a suite reporting only SKIP has tested nothing.

Case ID	Test case	Expected result	Status
TC-RLS-001	A student reads another student's worksheet submissions	0 rows returned	Not executed
TC-RLS-002	A student reads worksheet_item_keys (the answer key)	0 rows returned	Not executed
TC-RLS-003	A student reads another student's answers	0 rows returned	Not executed
TC-RLS-004	A student reads a task nobody assigned to them	0 rows returned	Not executed
TC-RLS-005	A student reads another student's task_assignees rows	0 rows returned	Not executed
TC-RLS-006	A student reads another user's notifications	0 rows returned	Not executed
TC-RLS-007	A teacher reads a task owned by another teacher	0 rows, or a documented exception for a teacher scheduled into the section	Not executed
TC-RLS-008	A teacher writes to teacher_subjects (admin only)	Insert refused	Not executed
TC-RLS-009	A student writes a notification addressed to a teacher	Insert refused	Not executed
TC-RLS-010	A student updates another student's submission	0 rows affected	Not executed
TC-RLS-011	Row Level Security is enabled on all 14 student-facing tables	All report "on"	Not executed
TC-RLS-012	No policy grants unconditional access (qual or with_check = true)	0 such policies	Not executed
TC-RLS-013	No table has RLS on with zero policies	0 such tables	Not executed

4. Functional test cases – automated

4.1 Unit and render tests

Source: src/lib/*.test.js and src/pages/dashboards/renderSmoke.test.jsx. Rows group the 62 assertions in these files by behaviour.

Case ID	Test case	Expected result	Status
TC-UNT-001	ITEM_TYPES lists the five supported question types	Exact list, in display order	Pass
TC-UNT-002	TRUE_FALSE_VALUES is exactly the two strings the answering UI uses	["True","False"]	Pass
TC-UNT-003	normalizePoints keeps a deliberate 0 as 0	0, not the 1 fallback	Pass
TC-UNT-004	normalizePoints falls back to 1 for blank, null, NaN and negative values	1	Pass
TC-UNT-005	splitAcceptedAnswers trims, drops blanks and never invents an answer	Clean list, or empty	Pass
TC-UNT-006	normalizeAnswer lowercases, trims and collapses internal whitespace	Normalised string	Pass
TC-UNT-007	Multiple choice scores the correct option and only that option	Full points / 0	Pass
TC-UNT-008	A blank answer counts as wrong, never as a match	0 points	Pass
TC-UNT-009	Identification accepts any answer the teacher listed, case-insensitively	Full points	Pass

Case ID	Test case	Expected result	Status
TC-UNT-010	Enumeration awards partial credit per matched answer, order-independent	Proportional points	Pass
TC-UNT-011	Enumeration counts a duplicated answer once and never exceeds item points	Capped at item points	Pass
TC-UNT-012	Partial credit rounds to two decimals	e.g. 1.67	Pass
TC-UNT-013	An item with an empty answer key returns nulls for manual scoring	Nulls, not zero	Pass
TC-UNT-014	scoreSubmission sums auto-scored points and reports the full total	score ≤ totalPoints	Pass
TC-UNT-015	An essay counts toward totalPoints but not toward score	Unscored, still counted	Pass
TC-UNT-016	formatCountdown reports days, hours, minutes and seconds as the deadline nears	Correct tone per band	Pass
TC-UNT-017	formatCountdown flips to “late” once the deadline passes	tone = late, isLate = true	Pass
TC-UNT-018	The exact deadline instant is not yet late	tone ≠ late	Pass
TC-UNT-019	combineDateAndTime falls back to the default time, and returns null with no date	Timestamp or null	Pass
TC-UNT-020	GRADE_LEVELS lists Grade 7 to Grade 12 in order	Six values	Pass
TC-UNT-021	normalizeGradeLevel accepts a bare number and differing case or spacing	Canonical “Grade N”	Pass
TC-UNT-022	canTeachSection allows a subject and grade the teacher holds	ok = true	Pass
TC-UNT-023	canTeachSection rejects the right subject at the wrong grade level	ok = false, reason given	Pass
TC-UNT-024	canTeachSection rejects a load belonging to a different school year	ok = false	Pass
TC-UNT-025	currentSchoolYear rolls over in June, not in January	“YYYY-YYYY”	Pass
TC-UNT-026	withRetry returns immediately on success, without sleeping	First result	Pass
TC-UNT-027	withRetry retries the configured number of times and returns the last result	Last result	Pass
TC-UNT-028	withRetry jitters the wait to 50–150% of delayMs	No synchronised retry storm	Pass
TC-UNT-029	Every student dashboard tab renders without throwing on first paint	No exception	Pass
TC-UNT-030	Every teacher dashboard tab renders without throwing on first paint	No exception	Pass
TC-UNT-031	Both rebuilt admin tabs render without throwing on first paint	No exception	Pass

4.2 Component interaction tests

Source: src/pages/dashboards/admin/tabs/TeachingLoadTab.test.jsx and SchedulesTab.test.jsx.

Case ID	Test case	Expected result	Status
TC-CMP-001	Teaching Load opens on Grade 7 and scopes its labels to it	“Subject for Grade 7”	Pass
TC-CMP-002	Pressing Add with nothing chosen names BOTH missing fields	Two red fields, two messages	Pass
TC-CMP-003	Add to list stages the entry and writes nothing	Draft shown, no DB call	Pass

Case ID	Test case	Expected result	Status
TC-CMP-004	Assign writes the whole draft in one call	addLoadEntries called once	Pass
TC-CMP-005	A failed write keeps the draft intact	Draft still on screen	Pass
TC-CMP-006	A teacher already holding something at this grade cannot be picked	Radio disabled	Pass
TC-CMP-007	A subject already taken at this grade cannot be picked	Option disabled, holder named	Pass
TC-CMP-008	Switching grade tab rescopes both the picker and the saved list	Grade 7 load absent on Grade 8	Pass
TC-CMP-009	Teacher search matches department, not only name	“mathemat” finds the Math teacher	Pass
TC-CMP-010	A failed teaching-load read offers Retry instead of an empty school	Error + Retry button	Pass
TC-CMP-011	Schedules shows only the chosen grade’s sections	Grade 8 sections hidden on Grade 7	Pass
TC-CMP-012	A section whose grade_level is spelled “7” still groups under Grade 7	Section visible	Pass
TC-CMP-013	Each section chip counts its filled subjects without being opened	“1/2 subjects”	Pass
TC-CMP-014	Nothing is offered to fill in until a section is chosen	“Pick a section above”	Pass
TC-CMP-015	An opened section lists every subject, filled or not	Unfilled rows say “nobody scheduled”	Pass
TC-CMP-016	A filled row reads back teacher, day, time and room	e.g. “Monday · 08:00–09:00 · Room 201”	Pass
TC-CMP-017	Add opens the form already knowing the section and subject	Both pre-filled	Pass
TC-CMP-018	The teacher dropdown offers only holders of that subject at that grade	Ineligible teacher absent	Pass
TC-CMP-019	When nobody is eligible, the form names Teaching Load as the fix	Guidance shown	Pass
TC-CMP-020	Create with no teacher marks the field and refuses to save	Red field, saveSchedule not called	Pass
TC-CMP-021	An end time not after the start marks BOTH time boxes	Two red inputs, one message	Pass
TC-CMP-022	Create saves once the form is valid	saveSchedule called once	Pass
TC-CMP-023	A failed schedules read offers Retry; an empty grade points at Sections	Distinct messages	Pass

4.3 End-to-end tests

Source: e2e/. Requires the seeded accounts and a running development server.

Case ID	Test case	Expected result	Status
TC-E2E-001	A student logs in and reaches the student dashboard	URL /student-dashboard, banner visible	Not executed
TC-E2E-002	A teacher logs in and reaches the teacher dashboard	URL /teacher-dashboard	Not executed
TC-E2E-003	A wrong password is refused and the user stays on the login page	No navigation	Not executed
TC-E2E-004	A student cannot get in through the staff login	No staff dashboard reached	Not executed
TC-E2E-005	A signed-out visitor cannot open a dashboard by typing its URL	Redirected away	Not executed
TC-E2E-006	Login completes in a sane time (guards against the auth deadlock)	Under 20 seconds	Not executed

Case ID	Test case	Expected result	Status
TC-E2E-007	The enrolment banner resolves instead of loading forever	No permanent "Loading your enrolment..."	Not executed
TC-E2E-008	Subject cards render and none of them is "Other"	Cards present, no Other card	Not executed
TC-E2E-009	A subject card opens the Tasks tab filtered to that subject	URL carries ?subject=	Not executed
TC-E2E-010	Every student tab opens without a blank screen	#root never empty	Not executed
TC-E2E-011	A failed stat read says so rather than showing a bare zero	Explicit failure text	Not executed
TC-E2E-012	The student notification bell opens	Dropdown visible	Not executed
TC-E2E-013	Every teacher tab opens without a blank screen	#root never empty	Not executed
TC-E2E-014	The worksheets tab distinguishes "no teaching load" from "read failed"	At most one message, correct one	Not executed
TC-E2E-015	The teacher notification bell opens	Dropdown visible	Not executed

5. Manual test cases

These write real data, so they are performed by hand rather than automated against the live database. Each is run by an admin, a teacher and a student account as the case requires.

Case ID	Test case	Steps	Expected result	Status
TC-MAN-001	Admin creates three sections for one grade level	Admin → Sections → add name, grade level, school year → Save	Sections appear, and each shows in the Schedules grade tab	Pending
TC-MAN-002	Admin enrolls students into a section	Admin → enrolment → assign 20 seeded students to a section	section_students rows exist; each student's banner names the section	Pending
TC-MAN-003	Admin assigns a teaching load in bulk	Admin → Teaching Load → Grade 7 → pick teacher + subject → Add to list ×8 → Assign	One toast reporting the count written; all eight appear in the grade's list	Pending
TC-MAN-004	Admin schedules a teacher into a section	Admin → Schedules → Grade 7 → section → subject row → Add → teacher, day, time → Create	Row appears under that subject with teacher, day, time and room	Pending
TC-MAN-005	Teacher creates a task and builds its questions	Teacher → Worksheets → New → title and type → add questions of each of the five types	Task saved with the teacher's own subject; item count matches	Pending
TC-MAN-006	Teacher distributes a task to specific students with a due date	Teacher → Worksheets → Distribute → section → tick some students → set due date and time → Distribute	Only the ticked students receive it; each gets a notification	Pending
TC-MAN-007	The task appears on the correct subject card for the student	Log in as an assigned student → Overview	The task is counted on the card named after the teacher's subject, never elsewhere	Pending
TC-MAN-008	The deadline countdown is live and turns late on time	Open the Tasks tab and watch the countdown across the deadline	Seconds tick; the badge turns red and reads "Late by ..." once passed	Pending
TC-MAN-009	A student answers and submits a task	Student → Tasks → Start → answer every item type → Submit	Status becomes Submitted; answers persist across a reload	Pending
TC-MAN-010	A late submission is recorded as late	Submit after the due date has passed	worksheet_submissions.is_late is true, stamped in Manila time	Pending
TC-MAN-011	The teacher is notified when a student submits	As the owning teacher, open the bell within one minute of a submission	A "New ... submission" notification naming the student and task	Pending
TC-MAN-012	Auto-checking scores the objective items	Teacher → Check Submissions → open a submitted worksheet	Objective items scored; essays left for manual marking	Pending

Case ID	Test case	Steps	Expected result	Status
TC-MAN-013	A score is invisible to the student until released	Before release, view the task as the student	No score shown anywhere, including in the browser's network responses	Pending
TC-MAN-014	Releasing a score makes it visible and updates Worksheet Performance	Teacher releases ¶ student reloads the Overview	Score visible; the performance percentage moves accordingly	Pending
TC-MAN-015	Opening the bell marks notifications read, and they stay read	Open the bell, close it, log out, log back in	The unread badge does not return	Pending
TC-MAN-016	A news post disappears from the news page after its expiry date	Admin sets an expiry date in the past ¶ open the public news page	The post is not listed, and the admin still sees it marked expired	Pending
TC-MAN-017	Three dashboards open at once remain usable	Log in as admin, teacher and student in three browser profiles at the same time	Each dashboard loads in a few seconds; no tab hangs	Pending
TC-MAN-018	A dropped connection shows a retry, never an empty list	Open a dashboard with the network throttled to offline, then restore it	An explicit failure message with a Retry button; never a silent "0" or empty state	Pending
TC-MAN-019	Light and dark mode are both legible on every rebuilt screen	Toggle the theme on Teaching Load, Schedules and the student Overview	No unreadable text; red validation fields visible in both	Pending
TC-MAN-020	A teacher cannot reach another teacher's data by URL	As teacher A, paste a URL containing teacher B's task id	No data from teacher B renders	Pending

6. Known limitations of this test suite

Stated plainly, because a test document that overstates its coverage is worse than none.

- The RLS and end-to-end suites have never been executed. The machine they were written on has no PostgreSQL client and could not download a browser binary. Their first real run is on the developer's machine.
- The render smoke test proves only that a tab does not throw on its first paint. It cannot see a button that does nothing.
- The Schedules smoke test renders only that tab's empty state, because its stub contains no sections. This was verified by removing an import used solely inside an opened section and watching the suite still pass; the component interaction tests exist to cover that gap.
- No automated test writes to the database. Distribution, answering, checking and releasing are covered by Section 5 only.
- No load or concurrency test exists. The 60-user trial is a manual exercise (TC-MAN-017).
- The teacher Grades tab uses a PostgREST embed with no foreign key behind it and will render an empty roster. Known, pre-existing, and out of scope for this round.